

Key issues on the TSI (so far)

Goal: to introduce students to the term structure and how it is applied in practise.

Key issues covered:

1. The features

- ✓ Definitions and how to build the term structure of interest rates. TSI
- ✓ Explaining the shape of the curve (2 theories)
 - ✓ yield curve shapes, shifts + drivers

2. Application in

- ✓ Bond pricing.
- ✓ Determining forward rates
 - Application of forward rates.
- ✓ Predicting economic conditions.

Exam questions: So far

Exam questions revolve around the following.

1. Calculating the bond price; deriving spot rate; forward^{fn} rates
2. Applying the spot rates (TSI), forward rates.
3. Explaining the shape of the curve and relating the curve future economic conditions.
a lot of Qs on shifts + drivers
4. The rest of the question requirements are side kicks e.g definitions, using those rates to work out some Chapter 14 calculations

“No matter how great the talent or efforts, some things just take time. You can’t produce a baby in one month by making nine women pregnant”

- *Warren Buffet*



Exam questions and curve balls

47, 40

4/10/2025 4/10/2026 4/10/2027 4/10/2028

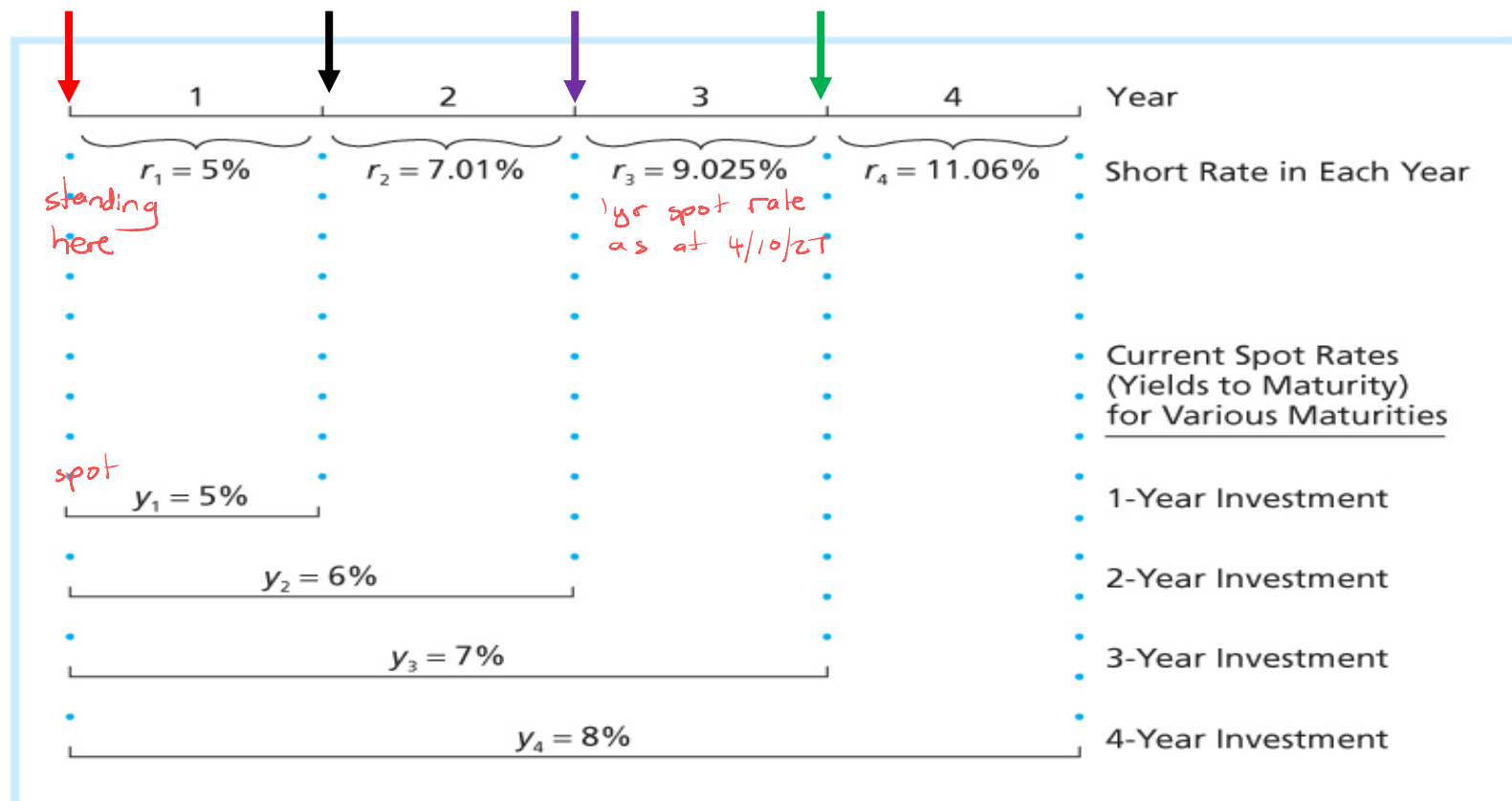


Figure 15.3 Short rates versus spot rates

- ❖ NB: exists today
- ❖ Spot or forward rate depends on your position.
- ❖ The spot rate is the geometric average of the current and future short rates; series of SHr that lie along its path
Know how to apply this [Price a bond using AFV given short rates]
- ❖ Assume the PET theory holds; determine 3-year spot rate after 1-year (ii) price of a 3-year ZCB after 1-year, etc geom ave of $r_2, r_3 + r_4$

Forward rate curve balls

- 1) What is the required rate of return (or interest rate or forward rate or return) in period two assuming the PET theory holds. *calc fwd rate*
- 2) Or assuming an environment of ^{no risk} certainty. *calc fwd rate*
- 3) Or assuming an environment of no risk. *calc fwd rate*

TSl-Revision

1. Determine the price of a corporate bond with the following features; Coupon rate=12% (paid once a year), time to maturity is 6years, face value =R1,000. From the bond market, government bonds similar to this bond are trading at a yield of 4%.

Inputs

❖ Cash flows

➤ Coupon = 12% x R1000 =
R120

➤ Face Value=R1,000

❖ YTM=4%+ Spread (e.g 2%)

❖ n=10

$$PV = R120 \times \frac{1 - \left[\frac{1}{(1.06)^{10}} \right]}{0.06} + \frac{R1000}{(1.06)^{10}}$$

$$**= R1,441.61**$$

❖ *Specify the 3 concepts I am testing*

❖ *Add a spread of your choice*

❖ *We will mark different final answers here*

TSI-Revision

You are given the following hypothetical TSI based on zero-coupon treasury bonds with a face value of R1,000

Maturity (years)	Yield to Maturity (%)
1	5%
2	6
3	7
4	8

Table 15.1

Prices and yields to maturity
on zero-coupon bonds
(\$1,000 face value)

Required

- ❖ You are given a 3-year bond issued by Vodacom Ltd, FV of R1,000, a coupon rate of 10% - paid annually.
- ❖ Required:
 - 1) Determine fair value of the Vodacom bond using the arbitrage-free valuation technique.

Time to maturity	Treasury spot rate	Adjusted spot rate for corporate bonds
1	5%	5%+1.5%=6.5%
2	6%	6%+1.5%=7.5%
3	7%	7%+1.5%=8.5%
4	8%	8%+1.5%=9.5%

$$Price = \frac{R100}{1.065\%} + \frac{R100}{1.075^2} + \frac{R1100}{1.085^3} = Answer$$

- ❖ Add a spread of your choice
- ❖ We will mark different final answers here

Know how to apply forward rates

Why is it important to understand forward rates

Why is it important to know forward rates.

1. to implement bond strategies.

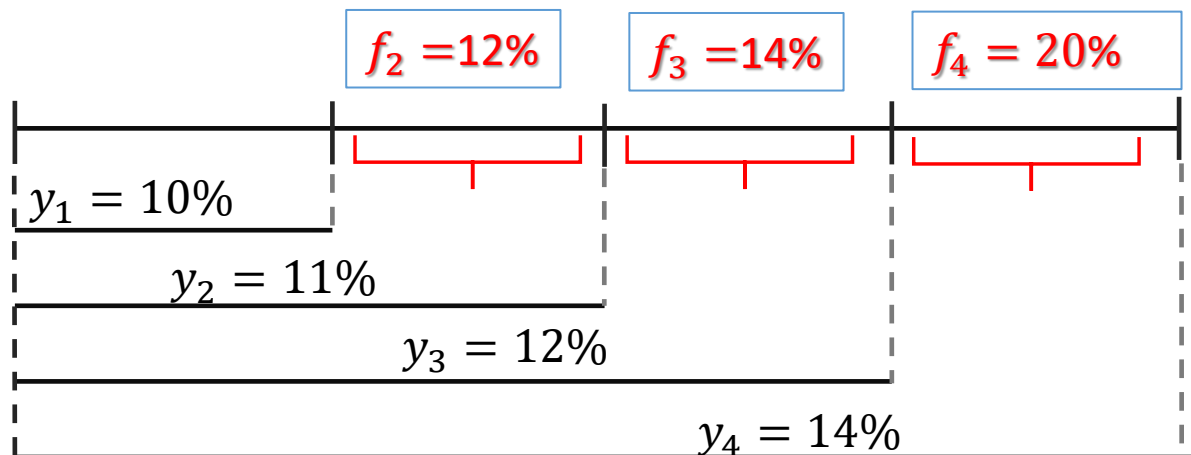
✓ **Alternative 1:** Buy a 2-year bond.

✓ **Alternative 2:** Buy a 1 year bond, and when it matures after 1year, buy another 1 year bond.

✓ **Decide whether to buy long/short term bonds**

2. Time your trade

3. Infer future interest rates & inflation



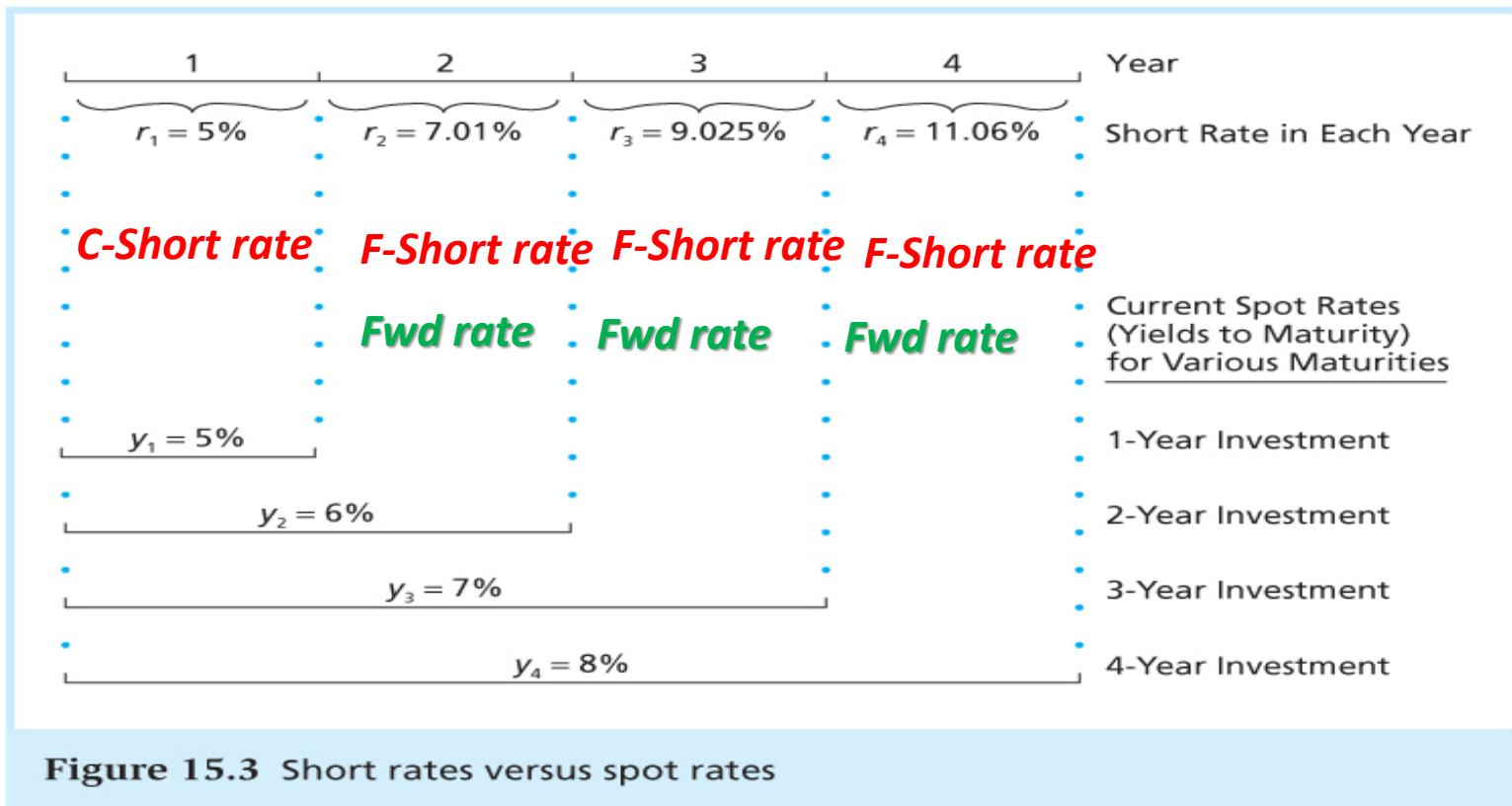
- ❖ How do you tackle a question that says, calculate the price of a 1-year ZCB with a face value of R1,000 after one year or after two years assuming:
 1. The Pure expectations theory holds
 - ~~2. Liquidity preference theory holds and long term investor demand a liquidity premium of 2%~~
- ❖ Calculate the 1 year holding period return of 3 year zero coupon bond.

$$\frac{P_1 - P_0 + C}{P_0}$$

$$P_0 = \frac{R1,000}{(1 + y_3)^3}$$

$$P_1 = \frac{R1,000}{(1 + y_2 @ \text{end of year 1})^2}$$

54:15



- 1)Calculate the 4-year RCY, assume the PET hold *f rates will exist in future ∴ C will be invested @ r_n*
- 2) Determine the 2-year spot rate after 2 years from now
- 3) Determine the 1-year holding period return of a 3 year bond, assume the PET holds [or assuming no risk or certainty

- ❖ Identify the shape of the yield curve from the information given in the table and interpret the curve assuming an environment of **certainty and uncertainty**. [4]

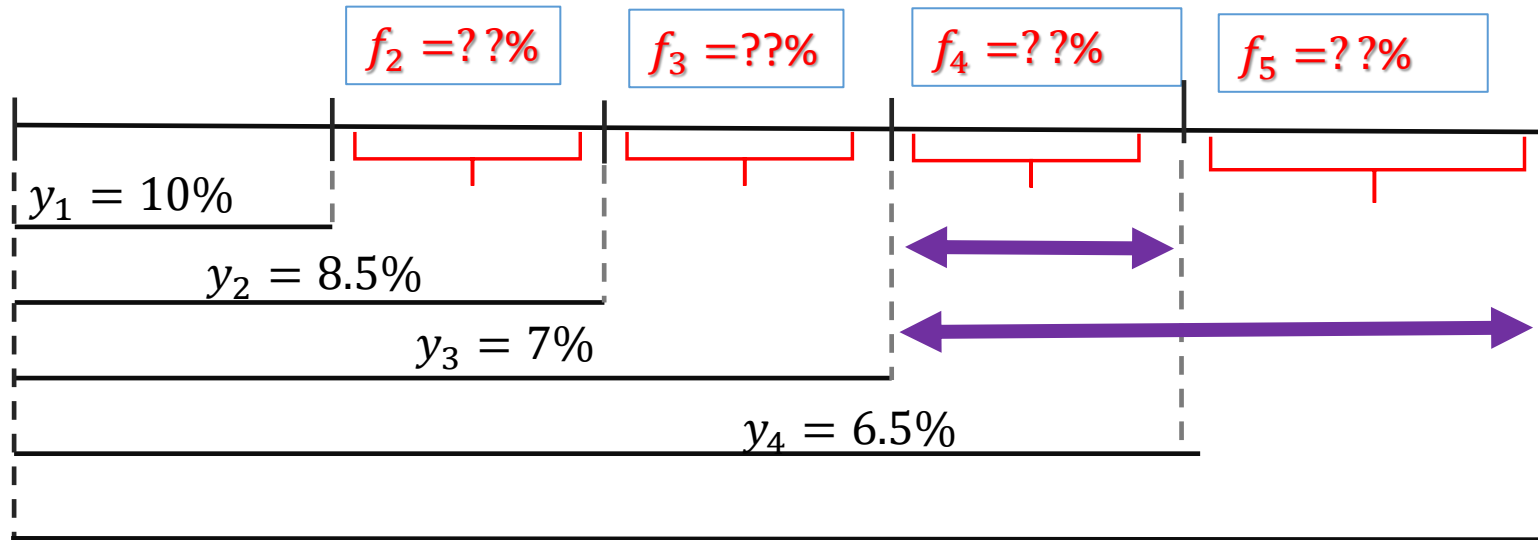
Test 1 Question

In the table below you are provided with the current and future expected prices of 1-year zero coupons bonds. All the bonds have a par value of R1,000. Today's date is 19th August 2022

Date	Price (R)
19 th August 2022	961.538
19 th August 2023	952.380
19 th August 2024	943.396
19 th August 2025	934.579
19 th August 2026	917.430

Calculate the arbitrage free value of a 4-year bond that pays a coupon of 6% per annum. The bond is currently trading at par, at a price of R880. **[10]**

went over this



- ❖ Assume that the PET theory is correct a) Calculate the 1-year and 2-year spot rates 2 years from now.

$$\diamond \text{ Spot rate (y)} = \left[\frac{\text{Maturity Value}}{\text{Price}} \right]^{\frac{1}{n}} - 1$$

❖ You are given the bond prices, the R1,000 amounts are face values.

Identify an inverted or upward sloping curve

3months	20-year	Spread between long & short term	Inverted or normal
2%	8%	6%	normal
1.5%	-2%	-3.5%	inverted
11%	4%	-7%	inverted